

Conditional Expectation and Linear Projection

M514: Econometrics II

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Expectation

Let $Y \in \mathbb{R}$ and $X \in \mathbb{R}^k$ be a random variable and a random vector of dimension k respectively. Whenever the joint density $f(y, x)$ of $(Y, X')'$ exists, the expectation function of $(Y, X')'$ is defined by

$$\mathbb{E}[(Y, X')'] = \int_{\mathbb{R}} \int_{\mathbb{R}^k} (y, x')' f(y, x) dx dy$$

The expectation operator $\mathbb{E}[\cdot]$ acts to integrate what is inside with the measure (weight) given by how likely it is for each value to be realized. We follow the convention to denote $f_X(x)$ and $f_Y(y)$ the marginal density of X and Y respectively, with the definition

$$f_X(x) = \int_{\mathbb{R}} f(y, x) dy$$
$$f_Y(y) = \int_{\mathbb{R}^k} f(y, x) dx$$

Expectation

Then, we have

$$\mathbb{E}[Y] = \int_{\mathbb{R}} y f_Y(y) dy$$
$$\mathbb{E}[X] = \int_{\mathbb{R}^k} x f_X(x) dx$$

For any measurable function $g : \mathbb{R} \rightarrow \mathbb{R}^l$, $h : \mathbb{R}^k \rightarrow \mathbb{R}^l$ with finite integer l , we also have

$$\mathbb{E}[g(Y)] = \int_{\mathbb{R}} g(y) f_Y(y) dy$$
$$\mathbb{E}[h(X)] = \int_{\mathbb{R}^k} h(x) f_X(x) dx$$

The last two equations are results of the *law of unconscious statistician*.

Variance

Variance of $X = (X_1, \dots, X_k)' \in \mathbb{R}^k$ is defined as

$$\begin{aligned}\text{Var}(X) &= \mathbb{E}[(X - \mathbb{E}[X])(X - \mathbb{E}[X])'] \\ &= \begin{pmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \dots & \text{Cov}(X_1, X_k) \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) & \dots & \text{Cov}(X_2, X_k) \\ \vdots & & \ddots & \vdots \\ \text{Cov}(X_k, X_1) & \text{Cov}(X_k, X_2) & \dots & \text{Var}(X_k) \end{pmatrix}\end{aligned}$$

Conditional Expectation

Whenever the joint density $f(y, x)$ of $(Y, X)'$ exists, the conditional expectation function (CEF) of Y given $X = x$ is defined by

$$m(x) := \mathbb{E}[Y|X = x] = \int_{\mathbb{R}} y f_{Y|X}(y|x) dy$$

where $f_{Y|X}(y|x)$ is the conditional density of Y given X given by

$$f_{Y|X}(y|x) = \frac{f(y, x)}{f_X(x)}$$

Note $\mathbb{E}[Y|X]$ is a function of the random vector X , and thus is a random variable. $\mathbb{E}[Y|X]$ is a random variable due to randomness in X . Note that randomness in Y has been integrated out by the conditional expectation operator.

Conditional Expectation

Law of iterated expectation:

If $\mathbb{E}|Y| < \infty$, then, for any random vector X ,

$$\mathbb{E}[\mathbb{E}[Y|X]] = \mathbb{E}[Y]$$

Proof:

Assume for convenience that the joint density of $(Y, X)'$ exists. Since $\mathbb{E}[Y|X]$ is a function of the random vector X , we have

$$\mathbb{E}[\mathbb{E}[Y|X]] = \int_{\mathbb{R}^k} \mathbb{E}[Y|X = x] f_X(x) dx$$

by the law of unconscious statistician.

Conditional Expectation

Thus,

$$\begin{aligned}\mathbb{E}[\mathbb{E}[Y|X]] &= \int_{\mathbb{R}^k} \mathbb{E}[Y|X = x] f_X(x) dx \\&= \int_{\mathbb{R}^k} \left(\int_{\mathbb{R}} y f_{Y|X}(y|x) dy \right) f_X(x) dx \\&= \int_{\mathbb{R}^k} \left(\int_{\mathbb{R}} y f_{Y|X}(y|x) f_X(x) dy \right) dx \\&= \int_{\mathbb{R}^k} \int_{\mathbb{R}} y f(y, x) dy dx \\&= \int_{\mathbb{R}} y \left(\int_{\mathbb{R}^k} f(y, x) dx \right) dy \\&= \int_{\mathbb{R}} y f_Y(y) dy \\&= \mathbb{E}[Y]\end{aligned}$$

where we've used $f_{Y|X}(y|x)f_X(x) = f(y, x)$ on the fourth equality, and Fubini's theorem on the fifth equality.

Conditional Expectation

Conditioning Theorem

$$\mathbb{E}[g(X)Y|X] = g(X)\mathbb{E}[Y|X]$$

Proof:

$$\begin{aligned}\mathbb{E}[g(X)Y|X = x] &= \int_{\mathbb{R}} g(x)yf_{y|x}(y|x)dy \\ &= g(x) \int_{\mathbb{R}} yf_{y|x}(y|x)dy \\ &= g(x)\mathbb{E}[Y|X = x]\end{aligned}$$

Conditional Expectation

From the law of iterated expectation and the conditioning theorem, we have

$$\mathbb{E}[g(X)Y] = \mathbb{E}[\mathbb{E}[g(X)Y|X]] = \mathbb{E}[g(X)\mathbb{E}[Y|X]]$$

CEF Error

We define CEF error e by

$$e := Y - m(X)$$

Tautologically, Y can be written as the sum of the conditional expectation $m(X)$ and deviation from it.

$$Y = m(X) + e$$

Exercise: Show that

$$\mathbb{E}[e|X] = 0$$

$$\mathbb{E}[e] = 0$$

$$\mathbb{E}[h(X)e] = 0$$

for any function $h(x)$ such that $\mathbb{E}|h(X)e| < \infty$.

CEF as the Best Predictor

Suppose the goal is to predict Y using a function $g(X)$ of X . Examples of possible g include the linear function¹,

$$g(X) = X'\beta = \sum_{i=1}^k \beta_i X_i$$

, and the quadratic function,

$$g(X) = \alpha + X'\beta + X'\Gamma X = \alpha + \sum_{i=1}^k \beta_i X_i + \sum_{i,j=1}^k \gamma_{ij} X_i X_j$$

Which g should be used for prediction? Notice that the prediction error $Y - g(X)$ is a random variable. One sensible criterion to evaluate a predictor g could be the expected value of the square of the prediction error.

¹In the linear case, we typically let $X = (X_0, X_1, \dots, X_{k-1})'$ with $X_0 = 1$ so that the intercept term is included without introducing new notation.

CEF as the Best Predictor

Suppose that we evaluate a candidate function g using the MSE criterion

$$\mathbb{E} [(Y - g(X))^2]$$

Then, $g(X) = m(X)$ is the best predictor in the sense that $m(X)$ minimizes the MSE.
To see this,

$$\begin{aligned}\mathbb{E} [(Y - g(X))^2] &= \mathbb{E} [(e + m(X) - g(X))^2] \\ &= \mathbb{E} [e^2] + \mathbb{E} [e(m(X) - g(X))] + \mathbb{E} [(m(X) - g(X))^2] \\ &= \mathbb{E} [e^2] + \mathbb{E} [(m(X) - g(X))^2] \\ &\geq \mathbb{E} [e^2] \\ &\geq \mathbb{E} [(Y - m(X))^2]\end{aligned}$$

This property suggests an (informal) interpretation of $\mathbb{E}[Y|X]$ as the projection of Y onto the space spanned by all functions of X .

Linear Projection as the Best Linear Predictor

The conditional expectation $m(X) = \mathbb{E}[Y|X]$ is the best predictor of Y among all functions of X . However, we typically do not know its functional form. Suppose we only consider a linear function $g(X) = X'\beta$ as candidate predictors. We define the **Linear Projection Coefficient** β^* as the minimizer of the MSE:

$$\beta^* = \arg \min_{\beta} \mathbb{E} [(Y - X'\beta)^2]$$

and the **Best Linear Predictor** of Y given X as

$$\mathcal{P}[Y|X] = X'\beta^*$$

Linear Projection as the Best Linear Predictor

The FOC of the minimization problem $\min_{\beta} \mathbb{E} [(Y - X'\beta)^2]$ is given by²

$$\begin{aligned} 0 &= \frac{\partial}{\partial \beta} \mathbb{E} [(Y - X'\beta)^2] \\ &= \frac{\partial}{\partial \beta} \mathbb{E} [Y^2 - 2YX'\beta + \beta'XX'\beta] \\ &= \mathbb{E} [-2XY + 2XX'\beta] \end{aligned}$$

which can be rewritten as

$$\begin{aligned} \mathbb{E}[XY] &= \mathbb{E}[XX']\beta \\ \implies \beta^* &= (\mathbb{E}[XX'])^{-1}\mathbb{E}[XY] \end{aligned}$$

Thus, the best linear predictor is given by

$$\mathcal{P}[Y|X] = X'(\mathbb{E}[XX'])^{-1}\mathbb{E}[XY]$$

²**FACT:**

$$\frac{\partial}{\partial x} b'Ax = A'b, \quad \frac{\partial}{\partial x} x'Ax = (A' + A)x$$

Linear Projection as the Best Linear Predictor

Define the linear projection error e as

$$e := Y - X'\beta^*$$

We have

$$\mathbb{E}[Xe] = 0$$

If we have $X_0 = 1$, this implies $\mathbb{E}[e] = 0$. However, we typically do not have $\mathbb{E}[e|X] = 0$. To see this,

$$\begin{aligned}\mathbb{E}[e|X] &= \mathbb{E}[Y - X'\beta^*|X] \\ &= \mathbb{E}[Y|X] - X'\beta^*\end{aligned}$$

which does not equal zero unless the true conditional expectation is linear.

Linear Projection as the Best Linear Predictor

Assuming the conditional mean zero assumption ($\mathbb{E}[e|X] = 0$) on the projection error e is equivalent to assuming a linear CEF function. To see this,

$$\begin{aligned} 0 &= \mathbb{E}[e|X] \\ &= \mathbb{E}[Y - X'\beta^*|X] \\ &= \mathbb{E}[Y|X] - X'\beta^* \\ \implies \mathbb{E}[Y|X] &= X'\beta^* \end{aligned}$$